

Differential Equations in Geophysical Fluid Dynamics

XXVI. Multi-layer and reduced gravity model

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Recap

Let us consider simplified primitive equation:

$$\frac{\partial u}{\partial t} - fv = -\frac{1}{\rho_0} \frac{\partial P}{\partial x} \quad (1a)$$

$$\frac{\partial v}{\partial t} + fu = -\frac{1}{\rho_0} \frac{\partial P}{\partial y} \quad (1b)$$

$$\frac{\partial w}{\partial t} = -\frac{1}{\rho_0} \frac{\partial P}{\partial z} - g \quad (1c)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (1d)$$

$$w|_{z=0} = \frac{\partial \eta}{\partial t} \quad (1e)$$

$$w|_{z=-h} = 0 \quad (1f)$$

Recap

Vertical averaging over entire water column yields the shallow water equation:

$$\frac{\partial \bar{u}}{\partial t} - f\bar{v} = -g \frac{\partial \eta}{\partial x} \quad (2a)$$

$$\frac{\partial \bar{v}}{\partial t} + f\bar{u} = -g \frac{\partial \eta}{\partial y} \quad (2b)$$

$$\frac{\partial \eta}{\partial t} + h \left(\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} \right) = 0 \quad (2c)$$

Layered ocean model

Let us consider two-layer density model:

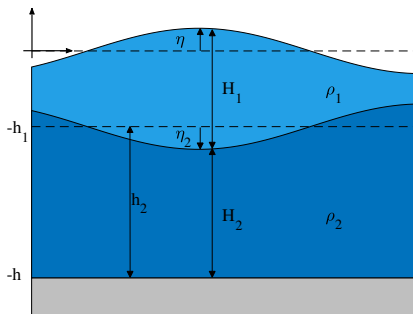
$$\frac{\partial u}{\partial t} - fv = -\frac{1}{\rho_0} \frac{\partial P}{\partial x} \quad (3a)$$

$$\frac{\partial v}{\partial t} + fu = -\frac{1}{\rho_0} \frac{\partial P}{\partial y} \quad (3b)$$

$$\frac{\partial w}{\partial t} = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g \quad (3c)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (3d)$$

$$w|_{z=0} = \frac{\partial \eta}{\partial t} \quad (3e) \quad w|_{z=-h_1} = \frac{\partial \eta_2}{\partial t} \quad (3f) \quad w|_{z=-h} = 0 \quad (3g)$$



Layered ocean model

Vertical averaging within each layer yields the layered ocean model:

≈ 1 (Boussinesq approximation)

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial x} \quad (4a)$$

$$\frac{\partial \bar{u}^{(2)}}{\partial t} - f \bar{v}^{(2)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial x} - g' \frac{\partial \eta_2}{\partial x} \quad (5a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial y} \quad (4b)$$

$$\frac{\partial \bar{v}^{(2)}}{\partial t} + f \bar{u}^{(2)} = -g \frac{\rho_1}{\rho_0} \frac{\partial \eta}{\partial y} - g' \frac{\partial \eta_2}{\partial y} \quad (5b)$$

$$\frac{\partial \eta}{\partial t} + h_1 \left(\frac{\partial \bar{u}^{(1)}}{\partial x} + \frac{\partial \bar{v}^{(1)}}{\partial y} \right) - \frac{\partial \eta_2}{\partial t} = 0 \quad (4c)$$

$$\frac{\partial \eta_2}{\partial t} + h_2 \left(\frac{\partial \bar{u}^{(2)}}{\partial x} + \frac{\partial \bar{v}^{(2)}}{\partial y} \right) = 0 \quad (5c)$$

where $g' = g(\rho_2 - \rho_1)/\rho_0$ is the reduced gravity.

Reduced gravity model

What if the subsurface layer is mostly motionless ($\bar{u}^{(2)} \approx \bar{v}^{(2)} \approx 0$)?

Subsurface momentum equation (5a) and (5b) become

$$\frac{\partial \eta}{\partial n} = -\frac{\rho_2 - \rho_1}{\rho_1} \frac{\partial \eta_2}{\partial n} \quad (6)$$

Using $H_1 = \eta + h_1 - \eta_2$, pressure gradient term in (4a) and (4b) becomes

$$\frac{\partial \eta}{\partial n} = -\frac{\rho_2 - \rho_1}{\rho_1} \frac{\partial H_1}{\partial n} \quad (7)$$

Continuity equation (4c) can be rewritten as

$$\frac{\partial H_1}{\partial t} + \frac{\partial(H_1 \bar{u}^{(1)})}{\partial x} + \frac{\partial(H_1 \bar{v}^{(1)})}{\partial y} = 0 \quad (8)$$

Reduced gravity model

So the governing equations of the two layer model are simplified to

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g' \frac{\partial H_1}{\partial x} \quad (9a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = -g' \frac{\partial H_1}{\partial y} \quad (9b)$$

$$\frac{\partial H_1}{\partial t} + \frac{\partial (H_1 \bar{u}^{(1)})}{\partial x} + \frac{\partial (H_1 \bar{v}^{(1)})}{\partial y} = 0 \quad (9c)$$

referred to as **reduced gravity model**. Is this mathematically different from the shallow water equations?